AGRICULTURAL ECONOMICS

"Agricultural economics is a science which deals the principles and methods of economics are applied to the special conditions of agricultural industry".

- It is derived from Greek word.
- Father of economics is ADAM SMITH.
- Chanakya called it as Arthashastra.

❖ **Four Important Definitions of Economics:**

- **Wealth definition:** By ADAM SMITH (father of economics)
  - ✓ An enquiry into the nature and causes of the wealth of nation.
- **Welfare definition:** By ALFERD MARSHALL
  - ✓ Study of mankind in the ordinary business of life examines part of individual and social action which is most closely connected with and use of material requisite of well-being.
- **Scarcity definition:** By LIONNER ROSBINS
  - ✓ Science which studies human behavior as a relation between ends and scarce means which have alternative uses.
- **Growth definition**: By J.M. KEYNES (father of modern economics)
  - ✓ Study of the administration of scarce resources and of the determinants of employment and income.

❖ **Contribution of Scientist:**

- **Adam smith:** Father of Economics, **Book "Wealth of nations"**
- **Alfred marshall**: Concept of law of diminishing marginal utility, law of equimarginal utility, consumer surplus.
- **J.M. Keynes:** Father of modern economics, Book "general theory of employment, interest, money.
- **Ragnar Frisch:** Terms "Microeconomics and macroeconomics"

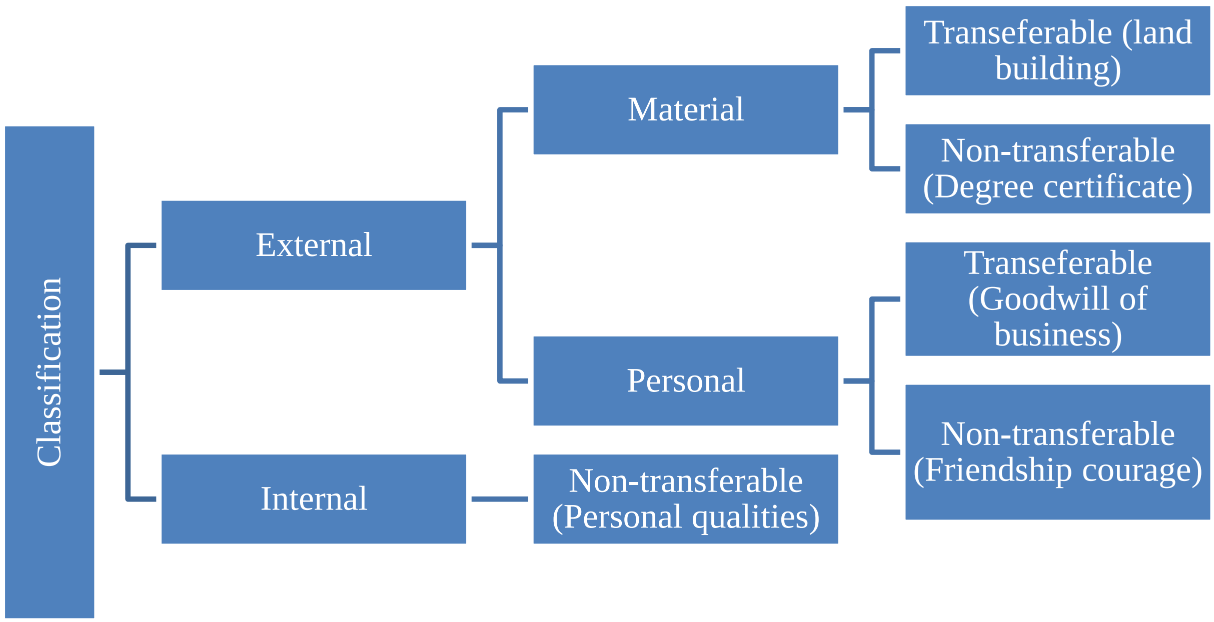
❖ **Economics Is Divided into Two Parts:**

A. **Micro-economics:** Such activities and services of consumption, production, exchange and distribution concerned with individual units viz., single industry, single farms, single consumer etc. i.e., at micro level are grouped under micro-economics.

B. **Macro-economics:** it deals with whole economic set-up and related additions or averages e.g., total production, total income, total employment, total expenditure, total savings, price level and economic development or whole economic setup.

❖ **Classification of Goods:**

- **Based on transferability**



- **Based on supply:**
  - ✓ Free goods (e.g., air): Have value in use
  - ✓ Economic goods: Have value in exchange and value in use.
- **Based on durability:**
  - ✓ Mono-period goods: That can be used only once.
  - ✓ Poly-period goods: That can be used multiple times.
- **Utility**: Want to satisfying power of a good is called utility.
- **Types of utility:**
  - ✓ **Form utility:** It is changing the form of goods i.e., processing a good.
  - ✓ **Place utility:** It is transportation of foods from one place to other.
  - ✓ **Time utility:** It is storage activity.
  - ✓ **Possession utility:** It is transfer of ownership of goods.

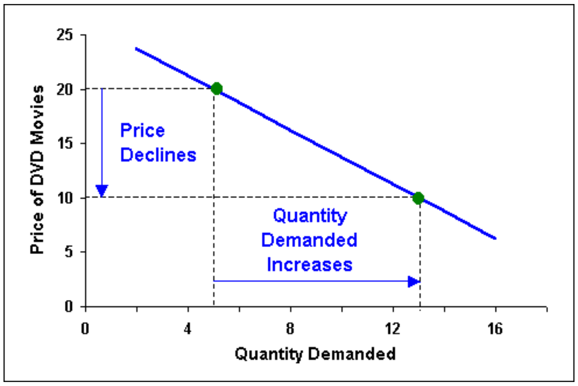
❖ **Consumer Behaviour:**

- **WANT:** It is just a wishful thinking (e.g., If you think of buying Range Rover).
- **DESIRE:** Person has ability to fulfil his/her want (e.g., If you think of buying purse, and you have money).
- **DEMAND**: Person has ability and fulfils his/her want at a particular place on a particular time (e.g., you bought a purse).

❖ **Law of Demand:**

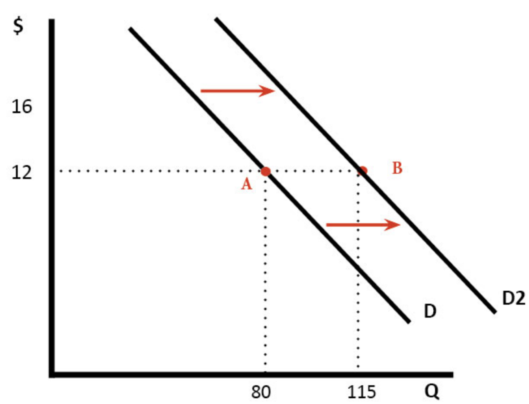
- When price of a commodity increase, quantity demanded of it will decrease and vice-versa.

➢ Fall in price leads to expansion in demand, while rise in price leads to contraction in demand.

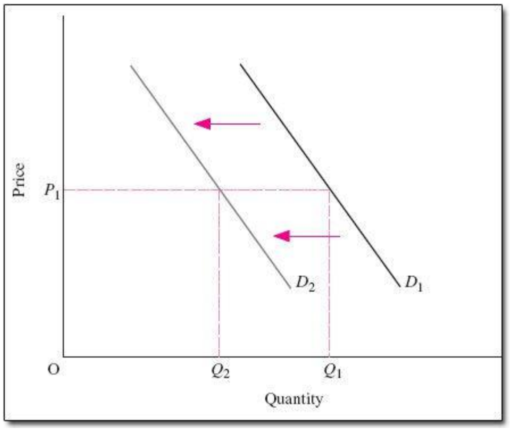


➢ Increase/ Decrease in demand i.e., shifting of demand: Depends upon other factors than price:
- ✓ Income.
- ✓ Substitute goods.
- ✓ Taste.
- ✓ Preferences.
- ✓ Fashion.
- ✓ Season.

a. Increase in Demand:



b. Decrease in Demand:



- **Autonomous demand:** No link with demand of other goods.
- **Derived demand**: Demand for certain goods related with demand for other goods. E.g., fertilizers, pesticides.

❖ **Kinds of Demand:**

- **Price demand:** Various quantity of goods that a consumer would be willing to buy at all possible prices in a given market at a given point of time.
- **Income demand:** Various quantity of goods that a consumer would be willing to buy at different level of income in a given market at a given pint of a time.
- **Cross demand:** Various quantity of goods that a consumer would be willing to buy not due to changes in price of commodity but due to changes in price related to goods.

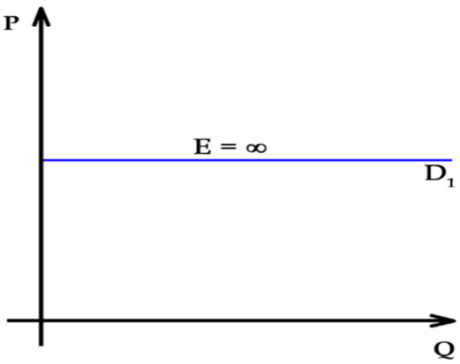
❖ **Giffen Goods: (Inferior goods)**

- It is an exception to law of demand.
- Demand of these goods generally decrease with decrease in price.
- Inferior goods are not giffen goods.

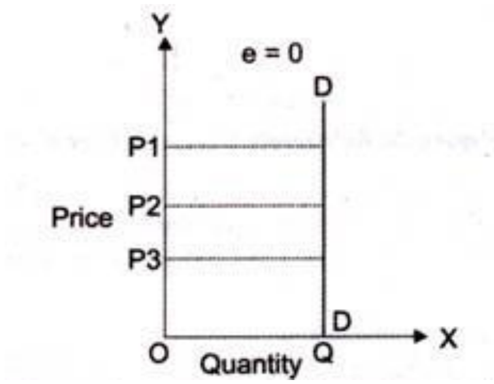
❖ **Elasticity of demand**: (Value ranging from 0 to ∞)

- Price elasticity of demand: $\frac{\%change\ in\ quantity\ demanded}{\%change\ in\ price}$
- Income elasticity of demand: $\frac{\%change\ in\ quantity\ demanded}{\%change\ in\ income}$
- Cross elasticity of demand: $\frac{\%change\ in\ quantity\ demmanded\ of\ good\ X}{(\%change\ in\ price\ of\ good\ Y)}$

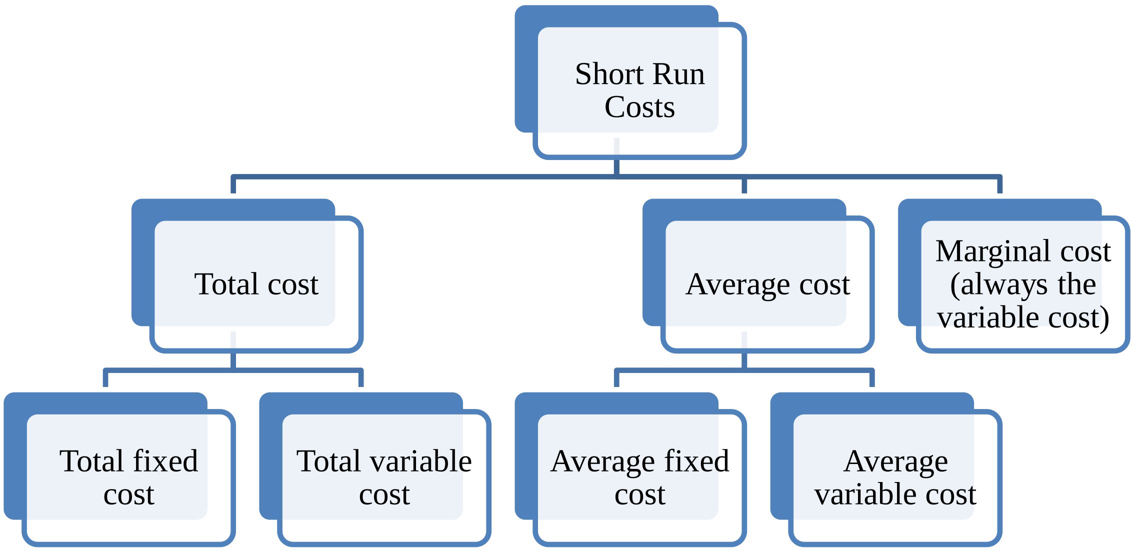
❖ **Perfectly Elastic Demand:**



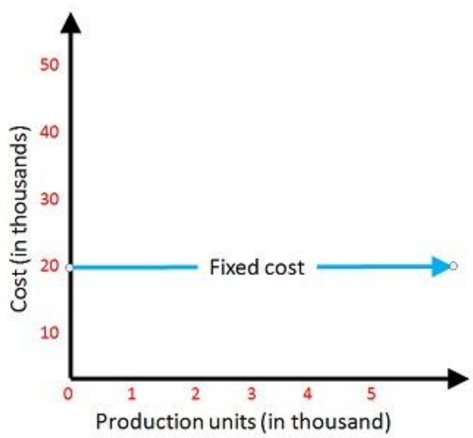
- **Perfectly Inelastic Demand:**



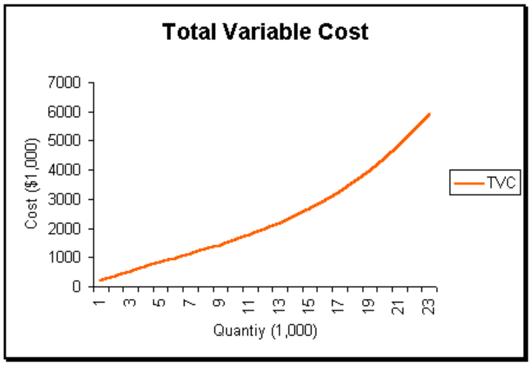
- **Cost Concepts**
  - In any business activities, the details of costs and returns provide an idea of profitability. Cost of production refers to the expenses incurred in producing a unit quantity of product in a particular time period.
  - Cost function can pertain to the short run or long run.
- **Short Run**: It is defined to be that period of time in which atleast one or some of the firm's inputs are fixed and some are variable.
- **Long Run**: It is defined to be time period in which all inputs are variable. There are no fixed inputs in the long run.



- ❖ **Fixed Cost:** Which do not vary with the level of output. Fixed costs are increased even in the absence of production.
  - ➢ E.g.: Taxes, Insurance, depreciation on machinery, tools, buildings etc.



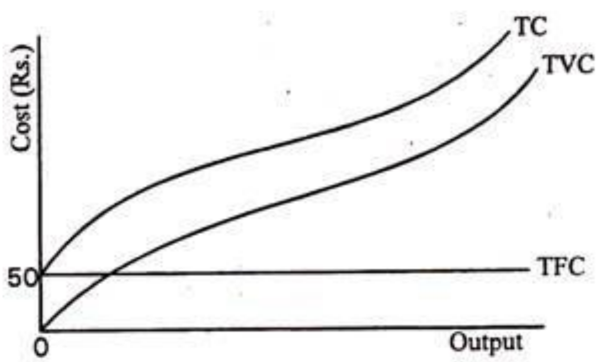
- ❖ **Variable Cost:** It varies with the level of output.
  - ➢ **E.g.:** Cost of raw materials, labour, power, repairs etc.
  - ➢ The summation of these costs refers to TVC.



❖ **Total Cost:**

➢ TC = TFC + TVC

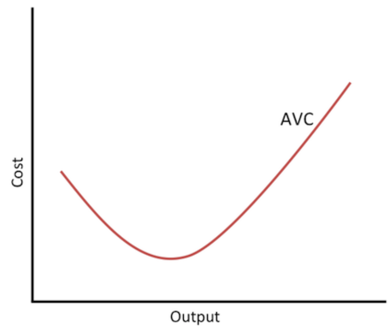
Units of Output	TFC	TVC	TC
0	10	0	10
1	10	10	20
2	10	18	28
3	10	24	34
4	10	28	38
5	10	32	42
6	10	38	48
7	10	46	56



❖ **Average variable cost**: It is the amount spent on the variable inputs to produce an unit of output.

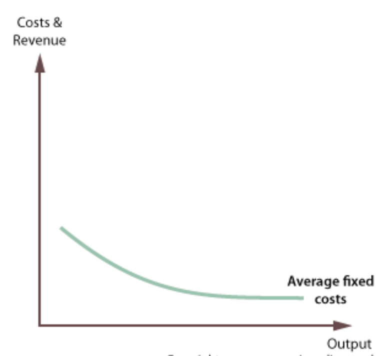
➢ It is total variable cost divided by output.

➢ $AVC = \frac{Total\ variable\ cost}{Output} = \frac{TVC}{Q}$



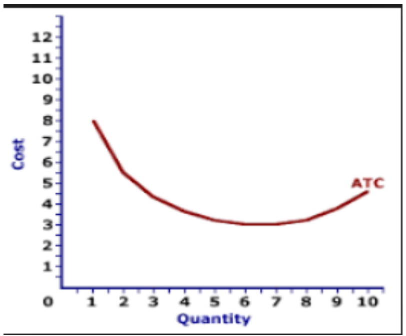
Units Output	TFC	TVC	TC	AFC	AVC	AC	MC
0	10	0	10	-	0	-	-
1	10	10	20	10	10	20	10
2	10	18	28	5	9	14	8
3	10	24	34	3.3	8	11.3	6
4	10	28	38	2.5	7	9.5	4
5	10	32	42	2.0	6.4	8.4	4
6	10	38	48	1.7	6.3	8	6
7	10	46	56	1.4	6.6	8	8
8	10	62	72	1.2	7.8	9	16

- **Average Fixed Cost (AFC):** It is the cost of fixed resources or inputs required for producing one unit of output.
  - $AFC = \frac{TFC}{Q}$
  - AFC curve is declining with the increased output because TFC is constant. Due to this it is continuously falling up to its maximum output.



- **Average Total Cost or Average Cost**: When the total costs are divided by output.

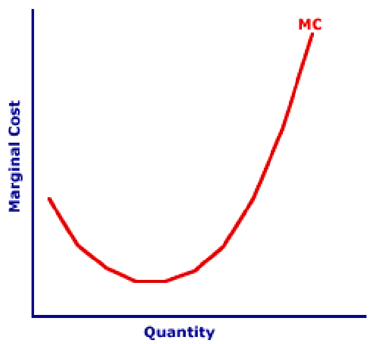
➢ $ATC = \frac{Total\ costs}{Output} = \frac{TFC+TVC}{Q}$



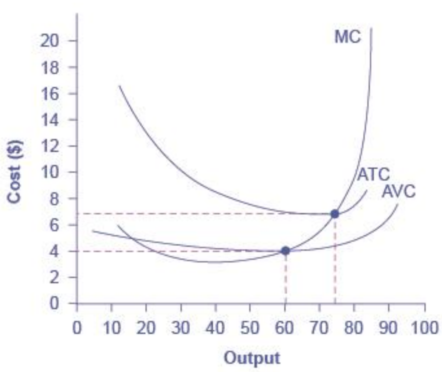
❖ **Marginal cost (MC):** It is the change in the total cost due to change in output.

➢ MC = $\frac{Change\ in\ Total\ costs}{Change\ in\ output} = \frac{\Delta TC}{\Delta Q} = \frac{\Delta TVC}{\Delta Q}$

➢ Note that, to compute MC, we can use TC or TVC because fixed cost can't be changed.



➢ **Imp:** MC curves intersect AVC and AC at their minimum points.



❖ **Important points determining price elasticity of demands:**

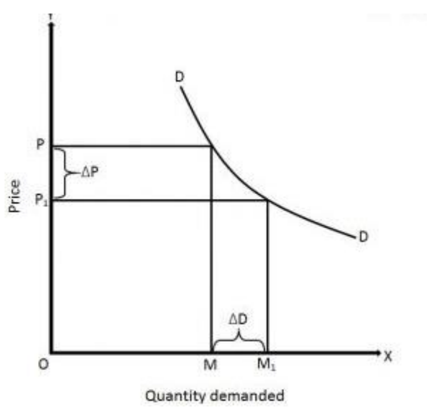
➢ **Nature of commodity:**

✓ Comforts, Ed = 1,

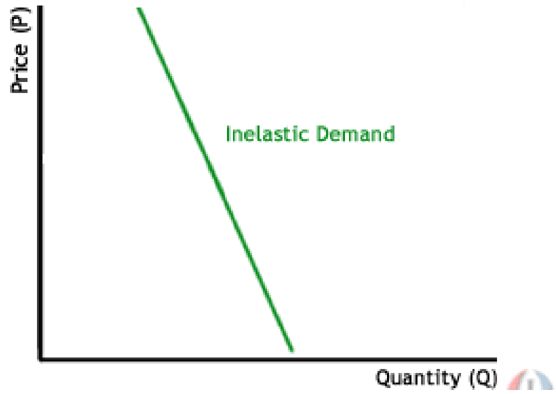
✓ Ed > 1

- ✓ Inferior goods, Ed < 1

- **Availability of substitute:**
  - ✓ If substitute available demand elastic.
  - ✓ If substitute not available inelastic.
  - ✓ It is so because, if price of one substitute fall people will buy more of it.
- **Goods with different uses:** These goods have elastic demand. E.g., electricity has many uses.
- **Postponement of use:** Goods whose demand can be postponed has high elastic demand like gold.
  - ✓ Whose amount can't be postponed has less elastic demand.
- **Income of consumer:** People who are very rich or poor has inelastic demand because rise or fall in prices has very little effect on their demand.
  - ✓ While middle class people have elastic demand.
- **Habit of consumer:** Demand for habitual goods is inelastic. Eg.: Cigarette, coffee (Despite rise in their price people demand such goods in same quantity).
- **Time:**
  - ✓ **In short period:** Demand inelastic.
  - ✓ **In long period:** Demand elastic.
  - ✓ Because a person can change his habits in the long run.
- **Relatively elastic demand:**

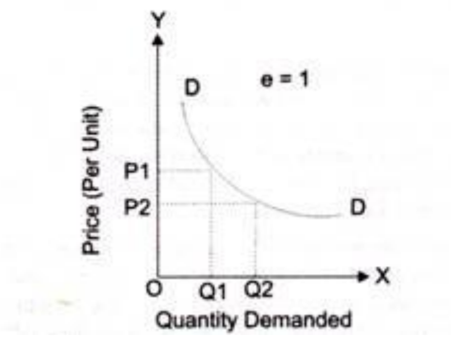


  - ✓ Ed > 1
  - ✓ Generally, for luxuries.
- **Relatively inelastic demand:**



- ✓ Ed < 1
- ✓ Generally, for necessities.

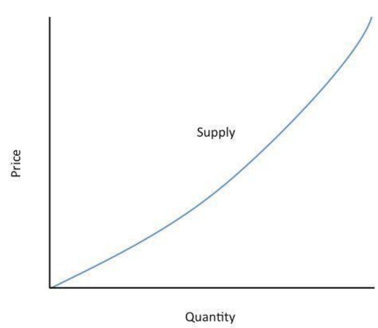
➢ **Unitary Elastic demand:**



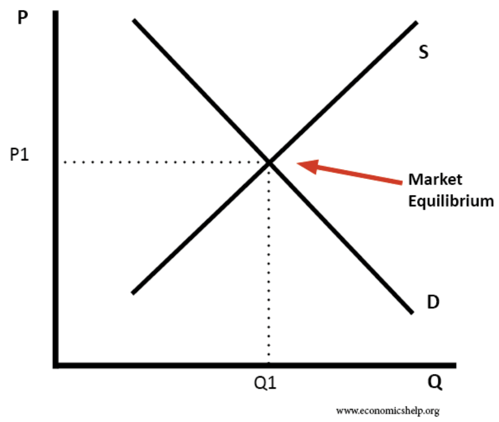
- ✓ Generally, for comforts.

➢ **Supply curves:**

- ✓ Production side
- ✓ Positive relation with price



❖ **Equilibrium condition:**



- ✓ Where demand + supply forces interest.

- ➢ **Rural Credit:** The word credit is derived from the Latin word Credo meaning "I believe" or I have confidence in". in economics, the term credit implies the postponement of payment. Farm finance is concerned with the acquisition and management of capital in the farm business.

❖ **Classification of credit:**

(i) **Short term credit:** It is given to meet current farm expenses such as purchase of seed, manure and fertilizers, payment of wages to hired human labour etc. and repaid in 12 months.

(ii) **Medium term credit**: it is given for purchase of land, construction of farm building, purchase of tractor and installation of tube wells etc. The period of long-term ranges from 12 month to 3-5 years.

(iii) **Long Term credit**: It is given to special projects and repayment period is more than 5 years.

- ➢ **Agriculture Finance :-** Credit is needed for adopting new technology, buying new machines, such as a machine which is beneficial to the farmer to increase production, but the farmer cannot buy it, in such a situation credit is needed.

  - ✓ As 54% of country's population dependent on agriculture/ production sector. Hence, finance is needed to regulate all these activities.
  - ✓ Agricultural Finance exist at both macro and Micro Levels.
  - ✓ Macro Finance deals with the different sources of funds for agriculture as a whole in the economy.
  - ✓ Micro Finance deals with the financial management of individual farm businesses..
  - ✓ **Finance**: Borrowed funds.
  - ✓ **Credit**: Owned funds and borrowed funds.

❖ **Cooperative Movement in India**

1. Initiation Stage (1904-11)
2. Modification Stage (1912-18)

3. Expansion Stage (1919-29)
4. Restructuring Stage (1930-46)

- **Co-operatives movement in India:**
  - The cooperative movement in India started in 1904 with the Cooperative Credit Societies Act.
  - The father of the cooperative movement in India, F. Nicholson.
  - In 1945, the government formed another cooperative planning committee. This committee submitted its report to the government in 1946.
  - The report of the All-India Rural Survey Committee was published in 1951. And the Committee chairman was A. D. Gorwala.

IMPORTANT INSTITUTES RELATED TO BANKING AND FINANCE:-

- **Reserve Bank of India :-** This bank was established on 1st April 1935 under the RBI Act, 1934 and it was nationalized on 1st January 1949.
  - It is the central bank of India.
  - The headquarter of Reserve Bank is in Mumbai.
  - **Functions of Reserve Bank :-**
    - ✓ **Issue the Notes** - The Reserve Bank issues notes of all denominations of India except one rupee coin or notes and small coins.
    - ✓ To act as a banker to the Government.
    - ✓ Acting as a banker to banks
    - ✓ Credit control.
    - ✓ Control over foreign exchange

- **Regional Rural Banks:-**
  - Regional Rural Banks were established in the country to provide institutional credit to the weaker sections of the society, especially farmers and artisans in rural areas.
  - Initially on October 2, 1975, five regional rural banks were established whereas at present 43 regional rural banks are functioning.
  - **Functions of Regional Rural Bank :-**
    - ✓ In this, there was a provision to provide credit and other facilities to small businessmen, agriculture, laborers, small and marginal farmers of rural areas.

❖ NABARD

- The National Bank for Agriculture and Rural Development (NABARD) was established on 12 July 1982. The paid-up capital of NABARD is 30000 crores
- NABARD is the apex body providing finance for agriculture and rural development in the country.
- NABARD, as an apex institution in the rural credit structure, provides refinance facilities to several financial institutions (State Land Development Banks, State Co-operative Banks, Scheduled Commercial Banks, and Regional Rural Banks), which provide wide spectrum of productive activities in rural areas. Give loans for promotion.
- NABARD receives funds from the Government of India, World Bank, and other agencies to fulfill its credit requirements.

INTERNATIONAL ORGANIZATION/INSTITUTIONS

❖ International Monetary Fund (Imf) :-

- **Establishment year :-** 22 July 1944
- **Work Starting :-** from 1 March 1947
- **Headquarter :-** Washington DC (USA)

❖ International Bank for Reconstruction and Development (IBRD) (World Bank)

- **Establishment year: -** July 1944
- **Work Starting :-** June 1946
- **Headquarter: -** Washington DC

❖ World Trade Organization (WTO)

- **Establishment year** In April 1994 (in the city of Marrakesh in Morocco on the agreement of the GATT Uruguay circle and duly established on January 1, 1995)
- **Headquarter :-** Geneva (Switzerland)
- **Work :-**
  - ✓ To provide facilities for the implementation, administration and operation of world trade agreements and multilateral and plural agreements.
  - ✓ To act as a forum for discussion among the members on any future matters relating to trade and tariffs.
  - ✓ To collaborate with the IMF and the World Bank for greater coherence in global economic policymaking.

❖ Asian Development Bank (ADB)

- **Establishment year:** - in December 1966
- **Work Starting :-** From January 1967
- **Headquarter :** Manila (Philippines)
- **Objective :-** Lending to Asian countries at concessional interest rates.

❖ Food and Agricultural organization (FAO)

- **Establishment year:** - 16 October 1945
- **Headquarter:** - Rome, Italy
- **Objective :-** To raise the standard of living and to improve the condition of rural people.
    - ✓ To increase the capacity of production and distribution of goods.
    - ✓ Its Director General is elected for 6 years.

❖ Indian Export Import Bank or EXIM Bank :-

- Export Import Bank of India was established on January 1, 1982. Its purpose is to provide financial assistance to exporters and importers.
- The headquarter of EXIM Bank is in Mumbai (Maharashtra).
- Exim Bank of India has overseas offices in Washington, Singapore, Abidjan, and Budapest (Hungary).

AGRICULTURAL MARKETING

- The word market comes from the latin word marcatus, which means merchandise or trade or a place where business is conducted. Word market has been widely and variedly used to mean (a) a place or a building where commodities are bought and sold, e.g., supermarket; (b) potential buyers and sellers of a product, e.g., wheat market and cotton market.
- "Marketing is the performance of all business processes, including the delivery of goods and services from the initial agricultural production point to the final (consumer) point."
- Agricultural marketing is a process in which the seller of agricultural commodities wishes to sell his goods and the buyer desires to buy that commodity at a fixed price.

❖ Utility:

- Marketing adds cost to the product; but, at the same time, it adds utilities to the
- product. The following four types of utilities of the product are created by marketing:

(a) Form Utility: The processing function adds form utility to the product by changing the raw material into a finished form. With this change, the product becomes more useful than it is in the form in which it is produced by the farmer.

(b) Place Utility: The transportation function adds place utility to products by shifting them to a place of need from the place of plenty. Products command higher prices at the place of need than at the place of production because of the increased utility of the product.

(c) Time Utility: The storage function adds time utility to the products by making them available at the time when they are needed.

(d) Possession Utility: The marketing function of buying and selling helps in the transfer of ownership from one person to another. Products are transferred through marketing to persons having a higher utility from persons having a low utility.

DIFFERENCE	PERFECT COMPETITION	MONOPOLISTIC	MONOPOLY
SELLERS	Large	Large	one
PRODUCT	Homogenous	Hetrogenous	Both
PRICE	Uniform	Non-Uniform	Non-Uniform
ENTRY OF FIRMS	Free Entry	Not Absolute Freedom	Not Possible
Knowledge of MARKET Condition	Perfect	Imperfect	Imperfect
Mobility	Perfect	Imperfect	Imperfect
Demand Curve	AR=MR	AR>MR	AR>MR
Selling Cost	Not Required	Often	seldom

❖ Classification of Agricultural Marketing:

(A) According to seller

1. Primary market
2. Secondary market
3. Tertiary market (Terminal market)

(B) According to time

1. Daily market
2. Short period market
3. Long period market

(C) According to competition

1. Perfect market
2. Imperfect market
3. Monopoly market

(D) According to regulation

1. Regulated market
2. Unregulated market

(E) According to function:

1. General or mixed market
2. Specialised market
3. Marketing by samples
4. Marketing by grades

(F) According to sale

1. General or mixed market
2. Specialised market
3. Marketing by samples
4. Marketing by grades

(G) According to quantity of product

1. Wholesale market
2. Retail market

A. According to seller:

1. **Primary market:** These markets are held once or twice in a week and are generally found in villages. Such markets are called Hats. In such markets producers directly sell their commodities to the consumers and itinerant dealers or village banias.
2. **Secondary market:** These are daily markets and are held in cities. Generally, wholesale of commodities takes place. Wholesalers, Kaccha and Pucca arhatiyas are present in secondary market.
3. **Terminal Market:** such markets are found at seashore or ports. The agricultural products are exported to the other countries by these markets.

B. According to time:

1. **Daily market:** Daily market is every short period market. The prices of agricultural commodities are determined on the basis of variation in demand. Supply is limited. Therefore, in equilibrium of demand and supply, the demand plays important role than supply.
2. **Short period market:** Due to short period the demand is more effective on supply. In short period market the supply of a commodity cannot be fully adjusted to the demand for it.

3. **Long period market:** in long period the supply of commodity can be adjusted to the demand for it. If the demand at a given price increases the production of commodity can be increased in the long period.

C. According to competitions:

1. **Perfect market:** In perfect market there are a large number of buyers and sellers and free competition among buyers and sellers. The buyers and sellers have full knowledge about the price of the commodities sold at different places.
2. **Imperfect market:** Buyers and sellers do not complete freely between themselves due to friendship. Sometimes the number of buyers and sellers are small.
   They do not know the exact price of the commodities sold at different places.
3. **Monopoly market:** When there is a single producer or seller of a commodity, and that commodity should not have any close substitutes it is said a monopoly market.

D. According to regulation:

1. **Regulated markets:** The markets are regulated under market regulation act. The representatives of government, cooperative societies, traders and farmers make a agricultural marketing society. These societies control all the marketing business. Middleman etc are given license and all the marketing charges are fixed and metric system of weighing is used.
2. **Unregulated markets:** Under such markets there is not any regulation of government. All the marketing defects are present in such markets. Middlemen charge more amount for their services.

E. According to area:

1. **Local market:** In such markets the buyers and sellers of a commodity carry on business at a particular place. The markets for perishable commodities such as milk and vegetables are largely local.
2. **National market:** In national market the buyers and sellers extend over the entire nation. The market of food grains and cloths are of national type.
3. **International market:** in such markets the buyers and seller are of different countries. The market for tea and jute is international.

F. According to function:

1. **General or mixed market:** the market in which articles of all sorts are offered for sale is called general market.
2. **Specialised market:** In specialized market, a particular commodity is sold, e.g., vegetable market, fish market etc.

3. **Marketing by samples:** The samples of all goods offered for sale are sent for display. For example, wheat, cotton, wool etc. are usually sold, by granding and sample. It is very difficult for the buyer to inspect every single grain of wheat that he buys. It is very difficult for a seller to transport all the goods to the marketplace. He there adopts the method of selling goods by sample.
4. **Marketing by grade:** Marketing by sample leads to marketing by grade, The commodity is divided into different grades on the basis of quantity or variety and each grade is given certain name or trademark. The sellers can sell the commodity without showing samples and purchasers can purchase the commodity by reference of name or mark of the grade.

G. According to sale:

1. **Wholesale market:** In such market wholesale marketing is done, Kacchha and pucca arhatiya, wholesales are transacting the commodities.
2. **Retail market:** In this market retail selling takes place. The consumers are directly in contact of retailers.

❖ Market/Marketing Channel:

A market channel is simply the path by which the commodity passes from producers to ultimate consumer or the manner in which the products is moved from one agency to another.

The nature of the market channel for any produce may very depending upon the region, the season of the year and the methods employed in handling. A single products may pass through different channels. The important channels are as:

1. Producer → Primary wholesaler→ Govt. procurement → Roll →Flourmill → Fair price shop → Consumer
2. Producer → Consumer
3. Producer → Village Bania → Kachha Arhatiya → Pucca arhatiya → Wholesaler → Consumer
4. Producer → dealer → Wholesaler → Consumer
5. Producer → Cooperative marketing society → Consumers
6. Producer → Kaccha Arhatiya → Pucca Arhatiya → Wholesaler → Miller → Retailer → Consumer

❖ Method Of Sale of Agricultural Produce:

1. **Under cover:** Under this system of sale the price is settled by brokers and arhatiyas under the cover of the cloth
2. **Open auction system:** Under this system the broker invites bid for the produce and the highest bidder is sold the produce. It is a fair system of sale.

3. **Sale by private agreement:** Under this system individual buyers make their offers. The decision is conveyed by the seller to the buyers.
4. **Dara sales:** In this system the heaps of grain of different qualities are sold at a flat rate.
5. **Close tender system:** Under this system tenders are invited from the buyers. The tender having the highest amount is accepted and commodity is given to him.

- **Producers Surplus of Agricultural Commodities:** From the marketing point of view, this surplus is more important than the total production of commodities.

1. **Marketable Surplus:** The marketable surplus is that quantity of the produce which can be made available to the non-farm population of the country. The marketable
   - surplus is the residual left with the producers' farmers after meeting his requirements.
   - MS = P - C (where MS = Marketable surplus P =Total production, and C = Total requirements)
2. **Marketed Surplus:** Marketed surplus is that quantity of the produce which the producer farmer actually sells in the market, irrespective of the requirements for family consumption, farm needs and other payments.
   - If marketed surplus is more than marketable surplus or farmer is selling against his wish, it will be known as distress sale.
   - Factors affecting marketable surplus: Size of holding, Production, Price of the Commodity, Size of family, Requirement of Seed and Feed, Nature of Commodity, Consumption Habits.

- **Marketing Costs:** The movement of products from the producers to the ultimate consumers involves costs, taxes, and cess which are called marketing costs. These costs vary with the channels through which a particular commodity passes through. Eg: - Cost of packing, transport, weighment, loading, unloading, losses and spoilages.
- **Reasons for High Marketing Costs:**

1. High transportation costs
2. Consumption pattern – Bulk transport to deficit areas.
3. Lack of storage facilities.
4. Bulkiness of the produce.
5. Volume of the products handled.
6. Absence of facilities for grading.
7. Perishable nature of the produce.
8. Costly and inadequate finance.
9. Seasonal supply.
10. Unfair trade practices.
11. Business losses.
12. Production in anticipation of demand and high prices.

13. Cost of risk.
14. Sales service

❖ Factors Affecting Marketing costs

1. Perish ability
2. Losses in storage and transportation
3. Volume of the product handled
2. Volume of the More – less cost
3. Volume of the Less – more cost
4. Regularity in supply : Costless irregular in supply – cost is more
5. Packaging : Costly (depends on the type of packing)
6. Extent of adoption of grading
7. Necessity of demand creation (advertisement)
8. Bulkiness
9. Need for retailing : (more retailing – more costly)
10. Necessity of storage
11. Extent of Risk
12. Facilities extended by dealers to consumers. (Return facility, home delivery, credit facility, entertainment)

FARM MANAGEMENT

- Farm management is made up of two words- **field** and **management**. "Farm" means an area used for cultivation by a farmer or a group of farmers.
- The meaning of management comes to change the production plan into action and adjust the goods used in it. So that more profit can be obtained.
- The study of field management helps the farmer to decide which crops to grow, how much to produce, how to produce, at what time to produce and which crops to sell and when to buy? etc., so that more profit can be obtained.
- **According to Adams,** "Farm management is the method of applying the principles underlying the selection, organization and operation of a farm industry. So that as much profit as possible can be obtained.
- Farm "Farm" is a specific area on which cultivation is done jointly by a farmer or a group of farmers. "
- **According to Tandon and Dhodhiyal,** "Packet management is a branch of agricultural economics, which studies the activities of earning and spending money by a farmer. Those concerned with the organization, operation and marketing of the farm.

OBJECTIVES OF FARM MANAGEMENT:

1. The main objective of farm management is to reduce cost, loss and damage, inefficiency and unemployment and to increase profit, productivity, efficiency and employment.
2. To study the resources obtained such as land, labour, capital, management, and pattern of production.
3. To get information about the relationship between different costs and receipts in carrying out agricultural works.
4. To know about the impact of new technology on the field.
5. Adopting crop rotation keeping in mind the size of the field, fertility, irrigation facilities etc.
6. To make proper distribution of resources among different occupations of the field.
7. To get information about the economy and economy of various industries etc.

❖ Three Main Points Arise in Farm Management:

1. How to get more production?
2. How to get more profit?
3. How to reduce the cost of production?

❖ Importance of Farm Management-

- Today agriculture is becoming a profitable business. One doe's agricultural work not only for his livelihood, but also to get profit. Therefore, the farmer selects only those crops and agricultural occupations in which maximum profit can be obtained, at present agriculture is done on commercial basis. Therefore, it has become necessary to study field management for the commercial and economic success of agriculture.

Principles of Farm Management :-

1. Principle of Variable Proportion (Law of Diminishing Returns)
2. Principle of Factor Substitution
3. Principle of Product Substitution
4. Minimum Cost Principle
5. Principle of Comparative Advantage
6. Principle of Opportunity Cost
7. Principle of Equi-marginal Returns

Economic Principles Applied to Farm Management:

The most important principle that help the farmers in making the choice and decisions to all phases of farming are:

1. The law of diminishing returns
2. Cost analysis
3. The law of comparative advantages
4. The law of equi-marginal returns
5. The law of substitution
6. Principle of opportunity cost
7. Principle of combining enterprises

- Farm management is a science which deals with the proper combination and operation of production factors including land, labour and capital and the choice of crop and livestock enterprises to bring about a maximum and continuous return to the most elementary operation units of farming. Farm management is concerned with decision making.
- **The main decisions are:**
  1. **What to produce:** It decides which products to produce. It shows the product-product relationship.
  2. **How to produce:** It relates to choosing the most effective method of producing a given quantity of a particular product. It shows the factor-factor relationship.
  3. **How much to produce.** It relates to the problems in converting the several farm resources into the final farm product. It shows the factor-prodct relationship
  4. **Time relationship:** It shows relationship between the time that farmer makes investment of capital and that later date when this investment returns the physical production

- These four relationships are based on physical or technical facts but require price and cost information in order to make economic interpretations.

Production and Factor of Production: -

- Production It is the process by which some goods and services, called inputs, are converted into other goods and services, called outputs, which create utility.

Production factors

- There are the four main factors of production:

(i) Land :- It is a main natural resource that generates income. It represents those natural resources which are scarce and profitable.

- **Features of Land:-**
  - ✓ This It is a gift of nature and is stable in quantity.
  - ✓ This is Permanent and not destroyed.
  - ✓ This is immovable and stable.
  - ✓ Geographical supply of earth cannot be increased, but economic supply can be increased. (By making the land denser and more used.)
  - ✓ There is considerable variation in the composition and fertility of Earth.

(ii) **Labour :-** When Any work is done by hands or by mind for the purpose of money, which is called labor.

- **Features of Labour**
  - ✓ Without help of labour Production cannot be done from the land.
  - ✓ Labour cannot be separated from the worker.
  - ✓ short term Labor supply cannot be adjusted with demand.
  - ✓ Changes affect labor cost and labor supply.
  - ✓ Labour is an active factor.

(iii) Capital :- The money which is used to create future wealth is called capital. Capital is not the real factor of production this is man made factor. All capital may necessarily be money, but not all money may necessarily be capital. All Capital is money, but not all money can be capital.

- **Features of capital:-**
  - ✓ Capital is a productive factor.
  - ✓ This generates income
  - ✓ Capital is a deactivating factor.

(iv) **Organization :-** Organization The main role of a business is to establish good relations and coordination with other companies.

- **Rewards for the Factors of Distribution**
  - ✓ Reward of Land - rent
  - ✓ Reward of Labour - wage
  - ✓ Reward of capital - Interest
  - ✓ Reward of business - Benefit

❖ **Some Basic Concepts:**

- **Farm:** Piece of land where livestock enterprises are taken under single management system and has specific boundaries.
- **Agriculture holding:** Total area of land owned by individual or joint family whether cultivated by family or rented out.
- **Operational holding:** Total area held under single management for the purpose of cultivation. It excludes any land leased out to another person.
- **Family holding:** Farm having gross income (Rs. 1600), net income (Rs. 1200).
- Optimum holding: $(3 \times family\ holding) \rightarrow maximum\ size\ of\ holding\ and\ family\ should\ posess.$
- **Economic holding:** Which provide a reasonable standard of living + give full employment.
- **Marginal farmer:** Having land less than 1 ha or 2.5 acre.
- **Small farmer:** Having land between 1 ha to 2 ha.
- **Fixed resources:** Resources which remain unchanged irrespective of level of production are called fixed resources (Exist in short run). E.g.: Land building.
- Variable resources: Resources which change with level of production are called variable resources, Higher the level of production, greater the use of resources and vice-versa. E.g.: Seed fertilizers.
- **Flow resources:** Resources which cannot be stored and should be used as and when they are available.
- **Stock resources:** Resources which can be stored, when they are not used in one production period. E.g.: Seeds, fertilizers, feed etc.
- **Cost of cultivation**: Expenditure incurred on all inputs and input services in raising a crop on a unit area written as wst in Rs/ha.
- Cost of production is expenditure incurred on providing a unit quantity of output is called wst of production written as Rs/tonne or Rs/quintal.

1. The law of diminishing returns:

- It is physical law of fundamental importance in organizing and operating the farm business. It is an important guiding factor in farming to decide the level at which a farmer can increase his output.
- It is practical experience of every farmer that after a certain stage in cultivation, the application of successive does of labour and capital to a given area of land does not bring about as much return as the previous does.
- Additional dose of labour and capital and labour applied to the cultivation of land causes in general, a less than proportionate increase in the amount produce unless in happens to coincide with an improvement in the arts of agriculture"

2. Cost analysis:

- This principle is of great importance in making decisions of farm business. Cost in general, refers to the expenses incurred on productive services and physical productivities are guided by the cost of various input factors, mainly costs are of two types:

(i) **Fixed cost:** Fixed costs present farming expenses of an overhead nature and donot change with the output, e.g., interest on fixed capital, depreciation, revenue etc.

(ii) **Variable costs:** it varies with the level of output. It includes changes of hired human labour, seed, manure and fertilizer, irrigation, insecticides and pesticides etc. These are seven costs of production

(a) Total variable cost (TVC): it is obtained by multiplying the amount of variable input by the price per unit of output.

(b) Average variable cost (AVC): it is computed by dividing total variable cost by the amount of output. AVC varies with the level of production.
Average variable cost is:
AVC = TVC/Outpur

(c) Marginal cost (MC): MC is the change in total cost as related to change in output. It is calculated by dividing the change in total cost by the corresponding change in output, i.e., Δ TC/ Δ Y. The marginal or added cost of each unit of output is important in determining how for we should push production and how much of the various resources we should use.

(d) Total fixed cost: total fixed cost shows the some of expenditures which will be incurred irrespective of the output.

(e) Average fixed cost: It is obtained by dividing total fixed costs by the amount output. As the output increase the average fixed costs decreases.
AFC = TFC/Output

(f) Total cost: It is the sum of total fixed cost and total variable cost.
TC = TFC + TVC

(g) Average total costs: It is worked out by dividing TC by output.
ATC = TC/Output

3. The law of comparative advantage:

- The law of comparative advantage helps to explain regional specialization. Farmers tend to produce those items which give more income at a lowest relative cost.
- With the income, they buy the items needed for production and for living which are produced at lowest relative cost in other areas.
- In short, specialised, and diversified farming are based on this principle. For example, sugarcane cultivation near sugar mill, vegetable cultivation near cities, rearing of sheep and goats in hilly areas will give more profit tha n the other places.

4. The law of Equi marginal Returns:

- Equi-means 'equal' marginal means additional or incremental. The law states that profit will be greatest if each unit of labour, capital and land is used where it adds the most to the return.

5. The law of substitution:

- The principle which helps selection from a number of alternatives is called law of substitution. According to this law the least cost combination of inputs or practices is obtained when the inputs or practices replaced is equal to the value of the input or the practice added.
- **Procedure for finding out least cost combination:**

(i) Computation of marginal rate of substitution (MRS) by dividing the number of units of the replaced input (X_1) by the number of units of added inputs (X_2).

Substitution ratio: = $\frac{no.of\ units\ of\ replaced\ input\ (x_1)}{No.of\ units\ of\ added\ input\ (x_2)}$

Where X_1 is replaced by X_2

(ii) Computation of price ratio by dividing the price of added input (Px_2) by the price of the replaced input (Px_1)

Price Ratio = $\frac{Price\ of\ added\ input\ (PX_2)}{Price\ of\ replaced\ unit\ (PX_1)}$

(iii) Calculation of least cost combination by equating (MRS):

MRS = $\frac{\Delta X_1}{\Delta X_2} = \frac{\Delta X_2}{\Delta X_1}$

- Least cost is realized at the point where the ratio of the input prices is inversely equal to their marginal rate of substitution.
- It is substitution ratio is greater than the price ratio, then we can lower costs by using more of the added input.
- If the substitution ratio is less than the price ratio then the cost can be reduced by using resource used in its most profitable alternative use.

6. Principle of combining Enterprises:

- An enterprises in farm business is defined as the production of a single crop or kind of livestock. Generally, farmers take more than one enterprise on their farms.
- The main aim of the farming is to get the maximum profit with minimum expenditure by combining enterprise.
- The combination of enterprises on a farm is influenced by the relationship that exists between the enterprises.
- The enterprises can have any one of the relationships explained below:

(i) **Independent enterprises:** These enterprises do not have direct bearing on each other. There is no effect on the other enterprises when level of one enterprises is influenced or decreased.

(ii) **Competitive enterprises:** Competitive enterprises are those enterprises which compete each other for the resources. Farmers have limited resources and therefore, all crop and livestock enterprises become competitive at some point. An increase in the output of one enterprise results in a decrease in the output of the other enterprises. Maize, bajra, paddy and sugarcane etc. in the same season compete with each other for resources.

(iii) **Supplementary enterprises:** Supplementary enterprises are those which do not compete for resources but help in increase income of the farmer. A small poultry enterprise is supplementary to other enterprise of the farm. A supplementary enterprise becomes competitive if it is expanded too far.

(iv) **Complementary enterprises:** Complementary enterprises are those enterprises which help each other in production and do not compete for resources. For example, if jowar is grown after berseem crop the yield of jowar increases because berseem fixes atmospheric nitrogen in the soil.

❖ Time Comparison Principle :-

- **Compounding**: It is calculating future value of the money held by a person at a present (by including rate of interest in it).
- **Discounting**: It is calculating present value of money if future sum of money will be given.

- **Principle of Opportunity Cost:-** The cost of the next best alternative foregone is called as Opportunity cost.
- **Principle of Equimarginal Returns :-** the limited resources should be allocated among alternative uses in such a way that the marginal value of the product of the last unit of the resource is equal in all uses.
- Minimum Loss Principle (Same as Cost Analysis in Microeconomics)

Types of Farming :-

- **Types of Farming** - When the classification of farming is done on the basis of use of land, size of farm, use of machines and machines, production of animals and crops, etc., then it is called type of farming, such as specialized farming, dry farming.
- **According to Dr. Ross,** "The type of farming refers to the use of land, the production and growth of crops and livestock, and the manner of agricultural activities."
  - ✓ Specialised Farming
  - ✓ Diversified Farming
  - ✓ Mixed farming
  - ✓ Dry Farming
  - ✓ Ranching
- **Specialized Farming:** Special farming means those holdings which get at least 50% of their income from any one enterprise or crop.
- In India, this type of cultivation is done only in the form of plantation in Assam or Nilgiri areas.
- **Diversified Farming:** In this there are many occupations of farmer's income. The farmer gets only a small part of his income from any one crop, less than 50%. Thus, there is no main source of income. Everyone gets some income. Diversified farming is also called general farming.

The Advantages of Multivariate Farming Are: -

1. Farmers remain independent.
2. The income of the farmer remains constant.
3. Animals, instruments, tools, land, etc., are all fully utilized.
4. To maintain soil fertility.
5. The family members continue to get work throughout the year.
6. The standard of living of the farmer is improves.

❖ Difference Between Specialized and Diversified Farming

Specialized	Diversified
One or two occupations are run on the farm	The number of occupations is many.
The main source of income is only one or two occupations.	There are many sources of income.
50% or more income is obtained from the main occupation.	The income of the farm is less than 50% from each occupation.
More information needed on specific occupation.	Information needed for different professions
Possible use of special types of tools and machines	Common farming methods.
Goods are often produced for sale in the market.	Often goods are produced for family consumption.
It is operated keeping in mind the principles of Industry.	Farming is done as a family need.

❖ Systems of Farming :-

- Peasant Farming
- Co-Operative Farming
- Capitalistic Farming
- Collective Farming (Russia and China)
- State Farming

❖ Depreciation

- The decline in the value of asset due to usage, accidental damage and time obsolescence.
- **Junk Value :-** The remaining value of an asset when it is completely destroyed. Or the value at the end of useful life.
- **Book Value :-** The value of an asset which is obtained by deducting the annual depreciation from the Original Cost every year.

❖ Methods of Depreciation

- **Straight Line Method**
  - ✓ Amount of Depreciation = (Original Cost of the Asset- Junk Value) / Useful life of asset.
- **Diminishing Balance Method**
  - ✓ The method assumes varying rates odf use of asset, year after year.
  - ✓ Amount of depreciation = (Book value of asset) X R

- ✓ Where R = Rate of Interest

- **Sum of the years digit method**
  - ✓ Annual amt. of depreciation =
  - ✓ (Original Cost- Junk Value) X Remaining years of useful life
  - ✓ Sum of the Digits from one through useful life
- **Annual Revaluation Method**
  - ✓ As the name indicates, the asset is revalued every year.
  - ✓ More specifically the value of an asset is calculated at the beginning and end of the year.
  - ✓ Generally, this method is used for land and livestock in agriculture.

❖ Farm Planning and Budgeting:

- Farm plan is the complete scheme for the operation and organization of a farm business. Farm planning is a process or technique for making plans about optimum choice and combination of farm enterprises.
- Farm planning serves the basis of farm budgeting, and a farm plan can be prepared without a budget.

❖ Factors effecting farm planning:

1. **Inventory of farm resources:** The inventory of farm resources should be taken at the beginning and the end of the year. It includes as appraisal of land, labour, capital (building, livestock, machinery, and layout) and management.
2. **Land:** The appraisal of the land will involve a determination of quality and quantity of the various types of soil. The location of land, types, irrigation facilities and fertility status are factors to be considered the value of land.
3. **Labour:** An examination of human and bullock labour programme will lead to a much effective organization of farm business.
4. **Capital:** It plays an important role in the farming business. Capital includes building, livestock, implement and machines and layout.
5. **Management:** It determines the success of business. It is a mental process and occurs when a farmer:
   - **(i)** observes and secures ideas,
   - **(ii)** analyses his observations and ideas,
   - **(iii)** makes decision on the basis of his analysis,
   - **(iv)** acts on the basis of his decisions and
   - **(v)** accepts the responsibility and consequences of his decisions.

❖ Partial Budgeting:

- It consists of a detailed description of the income and expenditure of an enterprise, a few enterprises or to a part of total farm.
- Symbolically it can be expressed as:

$R = Qy_1\ Py_1 - [X_1PX_1 = X_2PX_2 + \ldots\ldots\ldots\ldots.\ X_nP_n] - F.C.$

Where R = Net return

Qy_1 = Quantity of output

Py_1 = Price of output

X_nP_n = Inputs

P_x = Price of inputs

F.C. = Fixed cost

❖ Advantages of Partial Budgeting:

1. They are comparatively simple, quick and easy.
2. They may provide a clear indication of the profitability of a particular enterprise.
3. They provide a clear plant for utilizing a particular resource.
4. They may serve as an efficient means of adopting total budgets without compeletely reworking them.
5. By concentrating attention of a single enterprises or resource they are likely to result in fuller treatment of related problems.
6. They can be used to compare costs and returns associated with different practices such as of bullocks or tractor power.

❖ Complete Budgeting:

- It may be defined as detailed statement of the estimate of cost and returns of a farm as a complete unit.
- Symbolically it can be expressed as:

$R = [Q_{y1}P_1+Q_{y2}P_2+Q_{y3}P_3 \ldots\ldots\ldots\ Q_nP_n] - [X_1P_{x1}+X_2P_{x2}+ \ldots\ldots\ldots\ldots X_nP_n] - F.C.$

Were,

R = Net return of output

Qs = Quantity of output

Ys = Different enterprises

Ps = Price of output

X_n = Inputs

P_{xn} = Price of input

❖ **Advantages:**

1. It consists of the farm as a unit and all resources and enterprisses simultaneouly.
2. It may allow for substitution relationship between resources.
3. It allows for the complementary, supplementary, or competitive relationship between enterprises.
4. It drawn attention to the multitude of factors affecting farm earning.
5. It permits comparison of the probable total earning of the same farm with different combination of enterprises.
6. It avoids the necessity of allocating joint cost between enterprises.

❖ **Steps Involved in The Process of Budgeting:**

7. Making the inventory of farm resources (land, farm size, building, livestock, dead stock etc.)
8. Etimation of capital position of the farmer and determination of his credit needs.
9. Assessment of the availability and source of credit meet his credit requirement.
10. Determination of availability of family and hired human labour.
11. Working out cost and return in relation to the present operation so as to set objectives for future plans.
12. Drawing of alternate plans and settlement of the final plan in view of limitation.

❖ **Use of farm planning and budgeting:**

1. It is helpful in determination of needed adjustments as a yardstick for agricultural credit.
2. It is helpful in formulation and appraisal of development projects.
3. It provides road to farmer's security.
4. It provides basis for calculation of agricultural income of the whole country.

IMPORTANT FACTS:

- When Total utility Maximum, marginal utility is Zero.
- Elasticity of matchbox (necessities) is relatively non-elastic/inelastic.
- **Agriculture holding:** Total area of land owned by individual or joint family whether cultivated by family or rented out.
- **Operational holding:** Total area held under single management for the purpose of cultivation. It excludes any land leased out to another person.
- **Economic holding:** Which provide a reasonable standard of living + give full employment.
- **Marginal farmer:** Having land less than 1 ha or 2.5 acre.
- **Small farmer:** Having land between 1 ha to 2 ha.
- **Cost of cultivation:** Expenditure incurred on all inputs and input services in raising a crop on a unit area written as wst in Rs/ha.
- **Cost of production:** expenditure incurred on providing a unit quantity of output is called cost of production written as Rs/tonne or Rs/quintal.
- Law of Diminishing marginal utility is applicable to agriculture.
- Utility may be defined as the capacity of a good to satisfy human want. It is subjective and relative. Utility has no ethical and Moral significance. It is measured in terms of unit called **"Utils".**
- Transportation adds Place utility e.g., transportation of apples from J&K to other parts of country.
- Monopoly means: **Single seller**
- The warehousing corporation act came into: **18th March 1962**
- Some middlemen do not buy and sell directly but assist in the marketing process, they are known as- Facilitative middleman
- Storage adds time utility e.g., storage of wheat, rice to meet off season supply
- Processing adds from utility e.g., processing of wheat into flour and making juice from apple.
- An economy where allocation of resources is determined by the operation of price mechanism is termed as: **Market Economy**
- National income generated by agriculture is calculated by- **Output method**
- Marketing function of buying and selling adds possession utility e.g., selling of fruits, livestock.
- Inputs which are used to produce various goods and services – **Factors of production**.
- Anything above earth surface which is given by nature and is fixed- **Land**.
- APEDA is the coordinating agency for organic production and export under the brand name of -**Organic India**
- Capital goods that can be used for different purposes is called: **Floating capital**

- The markets dealing with agricultural commodities at district headquarter is called: **Secondary wholesale markets**
- For those goods having relative elastic demand the elasticity of demand will be: **more than 1**
- Law of diminishing returns is also known as: **Law of variable proportion**
- Price elasticity of supply of food products is generally- **<1**
- According to new rules after submission of application form in how many days the bank is supposed to provide KCC to farmers: - **2 Weeks**
- Preparing a farm budget in advance is known as farm planning
- The physical aspects of farm planning expressed in monetary term is called farm **budgeting**
- All capitals are considered as wealth, but all wealth is not necessarily capital. **E.g: farm** buildings are considered as capital and wealth whereas residential buildings are only capital.
- A person, who organizes, manages, taking a new idea (which always involves risks into it) and supervision ensuring success of business. – **Enterprise.**
- Anything that can satisfy a human want is called **"Good"**
- **"Services"** refers to the work that a person may do.
- Based on consumption:
  1. **Consumer goods –** Yield satisfaction directly. They are also called "Goods of the first order".
  2. **Producer goods –** These help to produce other goods known as the "Goods of the second order".
- The maximum quantity of output that can be produced from any given quantities of various inputs per period of time – **Production function (input output relationship).**
- The basis of farm budgeting is Benefit-**Cost analysis**
- The most commonly used economic tools for project evaluation is cost - **benefit analysis.**
- **Consumption** (Destruction of utility)
- **Production** (Creation/addition of utility
- **Distribution** (Sharing of all whatever produced)
- **Exchange** (Transfer of goods from one place to other)
- **Form utility:** It is changing the form of goods i.e. processing a good.
- **Place utility:** It is transportation of foods from one place to other.
- **Time utility:** It is storage activity.
- **Possession utility:** It is transfer of ownership of goods.
- **Want:** It is just a wishful thinking (e.g., If you think of buying Audi).
- **Desire:** Person has ability to fulfil his/her want (e.g., If you think of buying watch, and you have money).
- **Demand:** Person has ability and fulfils his/her want at a particular place on a particular time (e.g., you bought a watch).

- **Law of demand:**
  - When price of a commodity increase, quantity demanded of it will decrease and vice-versa.
  - Fall in price leads to expansion in demand, while rise in price leads to contraction in demand.
  - Autonomous demand: No link with demand of other goods
  - Derived demand: Demand for certain goods related with demand for other goods. **E.g., fertilizers, pesticides.**
- **Price demand:** Various quantity of goods that a consumer would be willing to buy at all possible prices in a given market at a given point of time.
- **Income demand:** Various quantity of goods that a consumer would be willing to buy at different level of income in a given market at a given pint of a time.
- **Cross demand:** Various quantity of goods that a consumer would be willing to buy not due to changes in price of commodity but due to changes in price related to goods.
- **Giffen goods:** (Inferior goods), It is an exception to law of demand.
- Demand of these goods generally decrease with decrease in price.
- Giffen goods are rare forms of inferior goods that have no ready substitute or alternative such as bread, rice, and potatoes
- e NAM was launched on **– 14th April,2016**
- The term **Oligopoly** means there are two or more sellers of a commodity.
- If the income elasticity of demand is greater than one, **the commodity is a luxury .**
- For consumer to be in equilibrium necessarry conditon is the ratio of Marginal Utility = prices.
- Annuity indicates the stream of payments (or) returns overtime.
- Capital in monetary form is called **Finance.**
- Capital owned from own savings is called – **Equity capital.**
- Capital owned from borrowed funds is called – **Debt capital**
- The ability of farm business to meet its debts or obligations to pay is called – **Solvency.**
- Minimum support price (MSP) for Agril. Commodities recommended by – **CACP (setup in 1965, recommended by L.K. Jha).**
- The area of operation of society is restricted to one village – **PACS.**
- **MSP** is price for maintaining Public distribution system by government.
- Demand for **Agril.** Produce is relatively more inelastic.
- Demand for necessary commodities is – **Inelastic**
- Demand for Luxurious commodities is – **More elastic.**
- Directorate of marketing inspection (DMI) is located at **Faridabad.**

- Elasticity of demand means degree of relation between quantities demanded/changes because of change in price.
- Elasticity of production refers to proportionate change in output as compared to proportionate change in input.
- The relationship between cost function and production function is – **Positive.**
- Indifference curve is always convex to origin (It is points of various combination of two commodities of same utility)
- Principle of equi-marginal returns is applied when the resources are limited.
- A positive science is concerned with **"What is"** (pure science) and normative science with **"What out to be"** (Normative Science).
- **"Economic laws"** are statements of uniformities which govern human behaviour concerning the utilization of limited resources for the attainment of unlimited ends – **Robbins.**
- Elasticity means sensitiveness (or) responsiveness of demand to the change in price.
- Elasticity of demand **= % change in demand/% of change in price.**
- When a small change in price may lead to a great change in demand. The demand is elastic. A big change in price is followed by only a small change in demand it is said to be in Inelastic demand **Eg: Salt.**
- Cros elasticity means a change in the demand for a commodity owing to a change in the price of another commodity.
- Perfectly elastic demand: Demand curve is horizontal to **'X' axis Ed = 1.**
- Perfectly in elastic demand: Demand curve is vertical straight-line **Ed = 0.**
- It the income elasticity of demand is greater than one, the commodity is a luxury.
- The relationship between marginal cost (MC) and fixed cost (FC): **MC is independent of FC.**
- **Farm:** Piece of land where livestock enterprises are taken under single management system and has specific boundaries.
- **Law of diminishing marginal return:** Law states that when we go on using more input for producing output. The production will increase initially at increasing rate, then increase at constant rate, then increase at decreasing rate, then seems constant and finally start decreasing at increasing rate.
- **Production function:** It is defined as non- mathematically and as well as graphical representation that describe the extent to which a particular product depends on quantity of input.
- **TPP:** Total amount of output obtained by using different unit of inputs, measuring in physical units like quintal, kg.
- **APP:** Reveals technical efficiency

- Factor/ factor Relationship/ Principle of factor substitution/ Principle of least cost combination:
- Deals with combination of inputs, so that the required level of production can be obtained with a Least Cost Combination.
- **Product-product relationship: -** Different combination in which 2 products are arranged in such a manner that they yield the maximum profit for the business. The main aim is profit maximization.
- **Types of products:** Joint product: Mainly in agriculture all products are joint products. Products which are produce from same production process are called joint products. **E.g.: Paddy, Straw.**
- **Complementary product:** If increase in one product cause increase in other product, when total quantity of inputs used on the 2 products are held constant. (don't compete for resources).
- Law of diminishing returns is also known as law of increasing costs or law of variable proportions.
- Law of diminishing returns (LDR) is the basic fundamental law of agriculture (To explain input-output relationship)
- Law of diminishing returns applies more generally to – **Agril. Industry.**
- Important Relationships and What they Helps in Deciding :-
- Factor- Product Relationship:- How much to Produce.
- Factor- Factor Relationship:- How to Produce
- Product- Product Relationship:- What to Produce
- According to LDR, optimum profit will be at point when **MC = MR**
- If MP > MC then further investment in production is profitable
- Farm management is the decision-making process.
- Objective of nationalization of banks were setup by- Indira Gandhi.
- District credit plant (DCP) is a blueprint of bankers containing technically viable and economically feasible scheme.
- In case of DIR scheme loans are advanced at **4%** interest per annum.
- Lead bank scheme was setup under chairmanship of – **F.K. Nariman**
- The number of banks nationalized so far are – **20 banks.**
- A banker to government of India is – **RBI (first governor of RBI is O. A. Smith)**
- Present farming system of India has become Market oriented.
- First regulated market in India was established in **– Karanjia cotton market (1886)**
- The resources which will remain unchanged irrespective of level of production- **Fixed resources** (found in short run) E.g.: land, farm buildings. Etc.

- ❖ The resources which will remain unchanged irrespective of level of production – Variable resources (Found in short run and long run) **e.g., seeds, pesticides etc.**
- ❖ Livestock insurance is included in **Variable Cost.**
- ❖ The total expenditure incurred for producing a unit of output using particular unit of input is **Cost of production (₹ per quintal).**
- ❖ The total expenditure incurred on all inputs used for producing of output per unit of land – **Cost of cultivation (₹ per hectare)**
- ❖ Isoquant means equal quantity or same amount of output.
- ❖ Inflation is calculated by the – **Consumer Price Index**
- ❖ During British period, government advances loans to the cultivators for production of crops, land improvement and distress relief are known as **Taccavi loans**
- ❖ Small amount of money loaned to a client by a bank – **Microcredit.**
- ❖ Nationalisation of 14 major commercial banks took place on – **19th july 1969.**
- ❖ Six more major commercial banks were nationalized during – **15th April 1980.**
- ❖ Regional rural banks (RRB) were established in **1975**.
- ❖ NABARD established as an apex agency for rural finance in **12th July 1982.**
- ❖ The term credit is originated from the Latin word 'Credo' or credere or creditum which means a loan or to entrust or I trust you.
- ❖ Agricultural finance corporation was promoted by – **Indian Banks Association (IBA).**
- ❖ Cooperative movement was started in India in – 1904.
- ❖ Father of cooperative movement in India – **F. Nicholson.**
- ❖ **Recommendation:**
    - ➢ Regional rural bank: Narsimhan committee
    - ➢ NABARD: Shivaraman committee
    - ➢ Single window system: Mohan kanda committee
- ❖ **Depreciation:-**The decline in the value of asset due to usage, accidental damage, and time obsolescence.
- ❖ **Junk Value:-** The remaining value of an asset when it is completely destroyed. Or the value at the end of useful life.
- ❖ **Book Value:-** The value of an asset which is obtained by deducting the annual depreciation from the Original Cost every year.
- ❖ **Methods of Depreciation :-**
    - ➢ Straight Line Method
    - ➢ Diminishing Balance Method
    - ➢ Sum of the Year digit method
    - ➢ Annual Revaluation Method
- ❖ India's first cooperative credit society was established in – **Karnataka**.

- The different between the demand for credit and supply is – credit worthiness.
- NABARD head office is located at – Mumbai.
- The headquarters of SBI is located – Mumbai.
- First RRB on pilot basis were setup on – 2nd Oct 1975(To give credit to Marginal, small and agril, labours).
- Single window system was started in – Andhra Pradesh.
- RBI was established on 1st April 1935 and nationalized in Jan 1st, 1949 (Present Governor - Shaktikanta Das).
- Price which necessarily covers variable cost and fixed cost is long run Price. Sub normal price is a short period price.
- The point where total revenue line and Total cost line intersect is known as Equilibrium price.
- The term market margin refers to profit of the middlemen.
- Scarcity rent arises due to high pressure on land.
- The line which represents various combinations of two inputs that can be purchased with given outlay of funds – Iso-cost line (Budget line, price line).
- Rental value of land is included in cost B.
- Farm is considered as socio-economic and decision-making unit.
- The ratio of output to input is called – **Technical efficiency.**
- **A line passing through least cost points on isoquant map is Expansion path**
- The line joining the endpoints of isoquant – Ridge line.
- The line by which all the least cost combination points are joint to each other – **Expansion path (isocline).**
- It will help in observing the availability e of cash in the business throughout the year-**Cash flow statement**
- In regulated market (ensure fair price) which one is not regulated – **price**.
- Village market is classified on the basis of – **Area**.
- Trading is in **secondary** wholesale market takes place between – **wholesaler and retailer.**
- Regular upswing and downswing occurring in prices during a year are called – **seasonal variations.**
- In the regulated market, the sale of the agricultural produce is undertaken by – **closed tender method.**
- **Hedging** practice is more common in foreign exchange market.
- Farm management is an intra farm science, farm planning helps in increasing farm income.
- Farmers obtain short term loans from **PACS**.
- Indian economy is mixed economy and Indian currency is convertible paper money.
- Fluctuation in agricultural prices is mainly due to shortage and surplus.
- A wholesaler assuming function of retailing in case of – forward marketing

- Harvesting is not a function of marketing.
- Secular markets are permanent in nature (manufactured goods such as TV, machinery).
- For perishable goods marketed surplus = **Marketable surplus.**
- Distress sale of marketing refers to selling price **< cost of production.**
- **Marketing channel** – The route through which agricultural products move from producers to consumers.
- A market guided by rules and regulations is **– Regulated market**
- In market, life blood is known as – **market information.**
- Marketing integration refers to expansion of firms by consolidating additional marketing functions under single management.
- Marginal product is the ratio of **Output/Input.**
- The first function performed in marketing of agricultural commodities is – Packing.
- The market is derived from Latin word marcatus which means trade or a place where business is conducted.
- The difference between the maximum price that consumers are willing to pay for a good and the market price that they actually pay for good is – consumers surplus.
- The quantity of produce which the producer farmer actually sells in the market, irrespective of his requirements – **Marketed surplus.**
- Marketable surplus is the residual left with the producer farmer after meeting his requirements for family consumption, farm needs, feeds for cattle etc.
- In case of perishable commodities, the marketed surplus = **Marketable surplus**
- The division of market into different homogenous group of consumers is **– Market segmentation.**
- Through all categories of farmers are interested in higher returns from their enterprises, what they really try to maximize is- **Net profit**
- When a new variety of crop is introduced- **partially budgeting is required.**
- A statement of assets and liabilities of a farm at a point of time **Balance sheet**
- Complete Budget refers to estimates of costs of returns from farm as a whole.
- The difference between the price paid by the consumer is willing to pay for a good and the market price that they actually pay for good is – **Consumers surplus.**
- The quantity of produce which the producer farmer actually sells in the market, irrespective of his requirements – **Marketed surplus.**
- The price at which government purchases food grain for maintaining the public distribution system for building buffer stock are known as- **Procurement prices**
- In our country Procurement Price is generally equal to the **Minimum Support Price.**
- While in normal situations Procurement Price must be greater than the **Minimum Support Price.**

- Marketable surplus – is the residual left with the producer farmer after meeting his requirement for family consumption, farm needs, feeds for cattle etc.
- **Marketable surplus = Total production (P) – Total requirements (C)**
- The division of market into different homogenous group of consumers is – market segmentation.
- **Market Structure:**
  - **Monopoly**: Market containing single seller of product
  - **Monopsony market:** Market containing single buyer of product
  - **Oligopoly market:** Market containing few sellers of commodity
  - **Oligopsony market:** Market containing few buyers of commodity
  - **Monopolistic market:** large sellers deal with heterogeneous and different forms of commodity
  - **Perfect competition:** large number of producers selling same product
- Complete Budget refers to estimates of costs of returns from farm as a whole.
- The difference between the price paid by the consumer and price received by the farmer is called – **Price spread.**
- **Balance sheet** provides net worth of business/ farm.
- Direct marketing is also known as – **farmers markets.**
- **AGMARK** is an indicator of purity.
- Agricultural produce (grading and marketing) act was passed in the year – **1937**.
- Who fixes grade standards for agricultural commodities in India – **Agril. Marketing advisor, GOI.**
- The time lag between investment and returns from agricultural projects is termed as – **Gestation period.**
- Crop loan system was 1st started in which state – **Andhra Pradesh.**
- Directorate of Marketing and Inspection (DMI) is located at **Faridabad Haryana**
- National Institute of Agricultural Marketing has started functioning at Jaipur (Raj) with effect from **8th August 1988.**
- Overhead cost is known as – **Fixed cost**
- Price theory is a branch of – **Microeconomics**
- Average cost is equal to – **Total cost / Output**
- The market used for food grains: **Regional or State market**
- The market used for durable goods: **National market**
- A time-based market basically for perishable goods: **Short period market**
- Food grain markets, vegetable markets, wool market are the example of: **Special market**
- Which market ensure fair price: **Regulate market**
- The market which is permanent in nature: **Secular market**

- Land rent is an example of **fixed cost**
- Utility has four type: Form Utility, Place Utility, Time Utility and Possession Utility
- Prime cost is also known as – **Variable cost**
- The 1st function performed in the marketing of agricultural commodities: **Packing**
- The Warehousing Corporations Act came into operation on 18th March 1962
- opportunity coast is also known as – **alternative cost**
- Diminishing return is also known as – **Law of variable**
- The most effective way to overcome the defects of agricultural marketing is – **Regulating marketing**
- The law of variable proportion is generally referred to as the – law of diminishing return production is a function of – **Factors**
- The relationship between Marketable to Marketed surplus for perishable products: **Equal**
- MSP is recommended by: **Commission of Agricultural Cost and Price (CACP)**
- MSP is fixed by: **GOI (CCEA)**
- The difference between value in use and value in exchange: **Consumer surplus**
- In mixed farming, the contribution of livestock to gross farm income is at least
- Input -Output relationship is also known as – **production function**
- Those products, in which two products produced together are called – **joint product**
- Dumping activity is seen under – **monopoly**
- Increase in money supply and fall in production causes: **Inflation**
- Export – Import bank of India was set up on – **1st January 1982**
- IRDP was stared in the year of – **2 October 1980**
- When marginal production is zero, then total production is – **maximum**